COMSATS University Islamabad, Lahore Campus

COAL Lab Assignment 03

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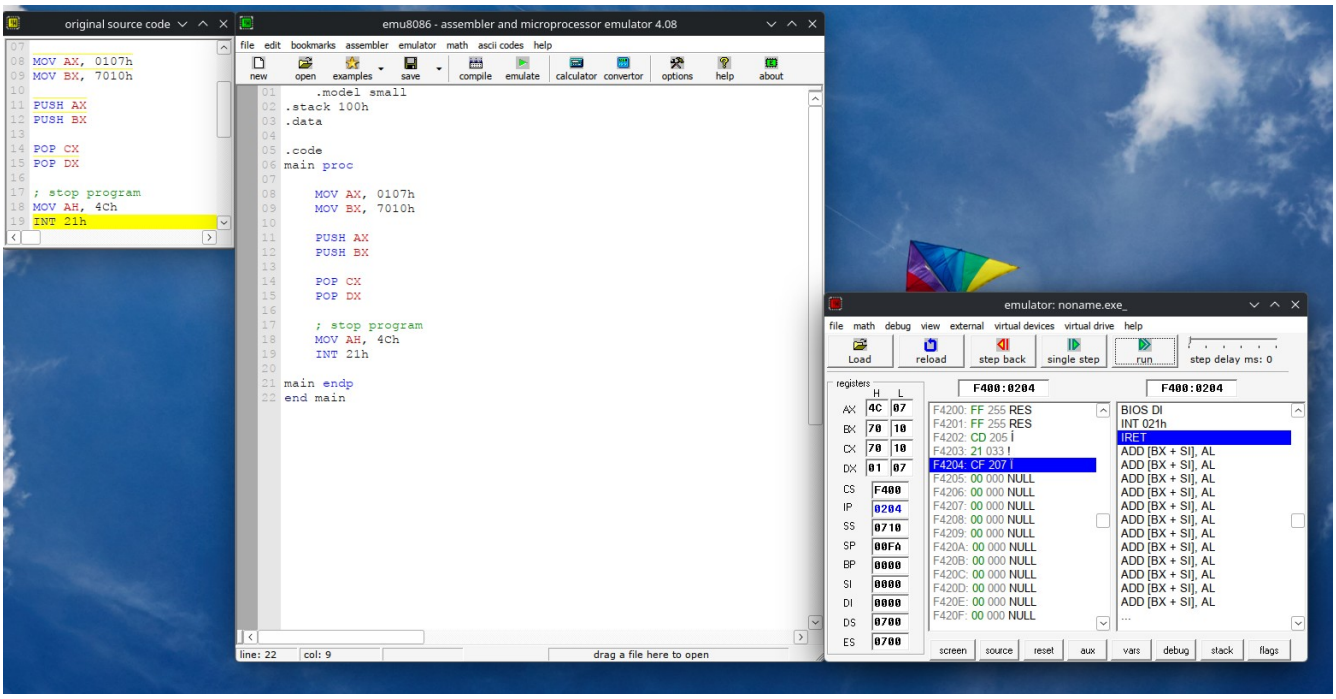
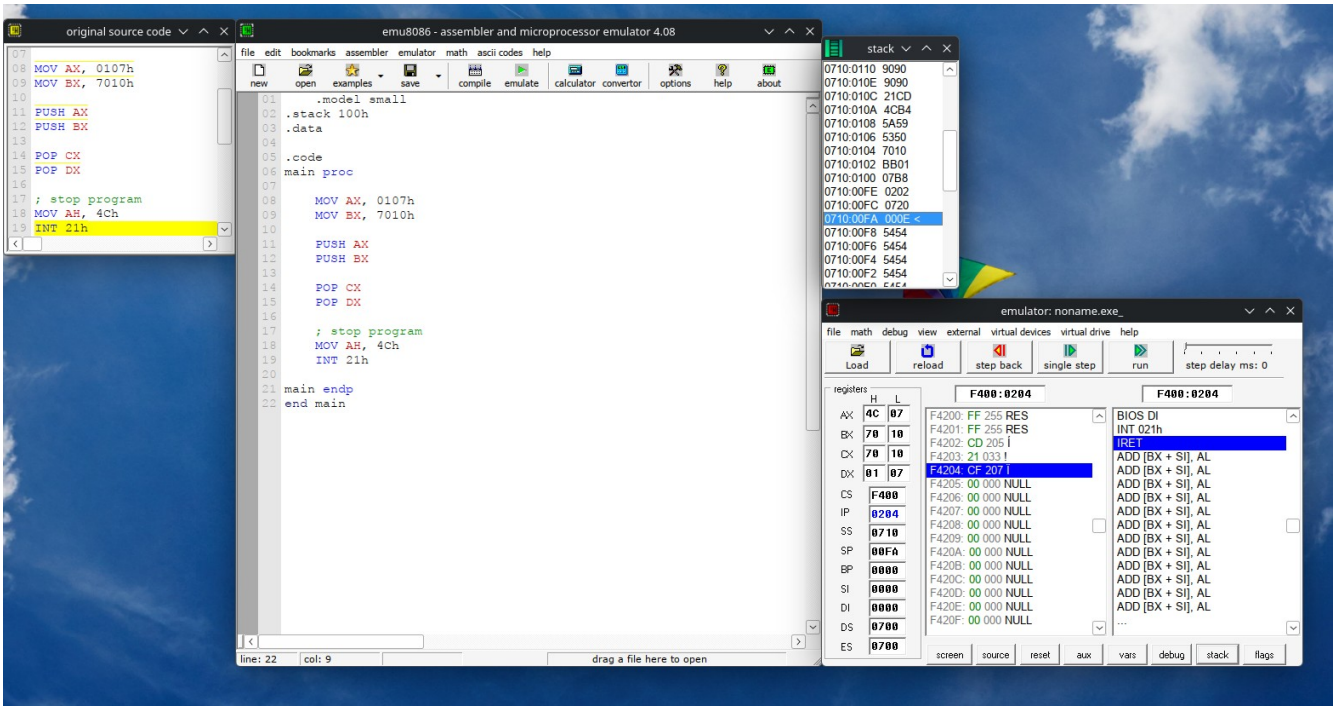
Submitted by: Samiullah Ashfaq

SP24-BSE-107-A

A. Stack Operations

```
.model small
.stack 100h
.data

.code
main proc

    mov ax, 0107h       ; RollNumber
    mov bx, 7010h       ; Reverse RollNumber

    push ax
    push bx

    pop cx
    pop dx

    mov ah, 4ch
    int 21h

main endp
end main
```

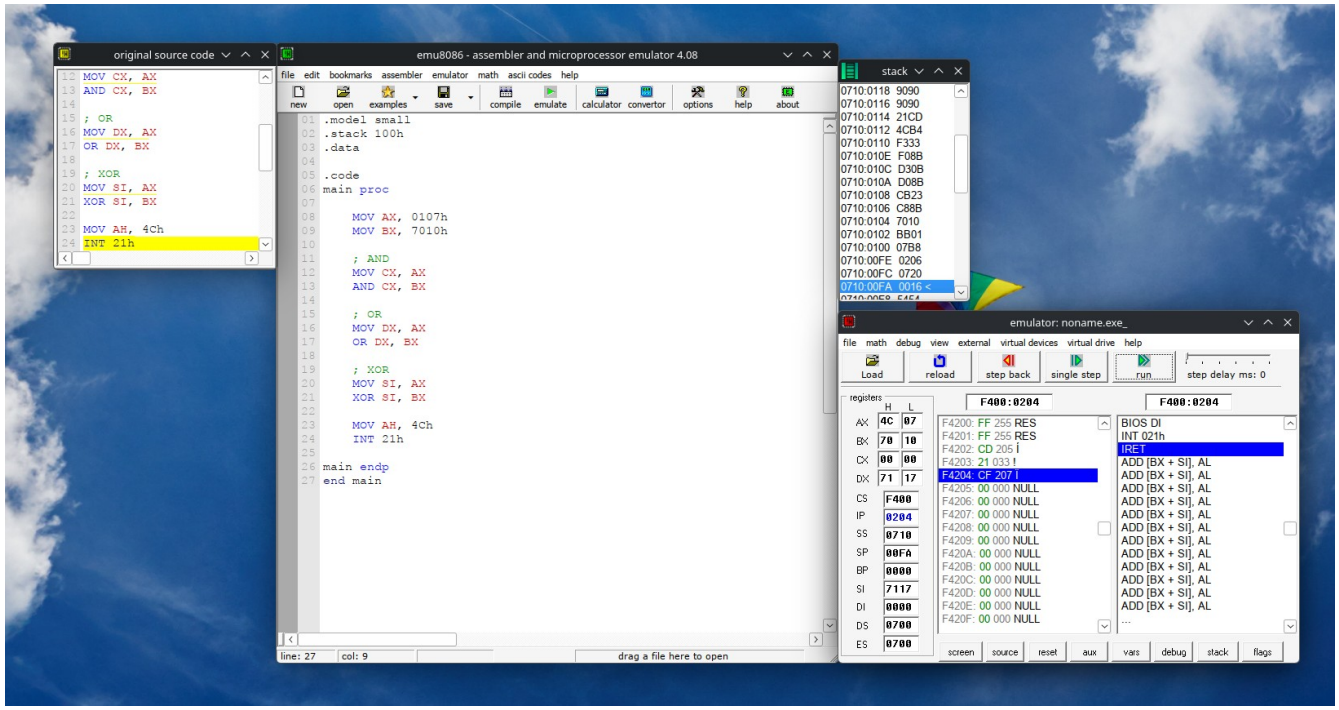




B. AND, OR, XOR

```
.model small
.stack 100h
.data

.code
main proc

    MOV AX, 0107h
    MOV BX, 7010h

    ; AND
    MOV CX, AX
    AND CX, BX

    ; OR
    MOV DX, AX
```

```
    OR DX, BX

    ; XOR
    MOV SI, AX
    XOR SI, BX

    MOV AH, 4Ch
    INT 21h

main endp
end main
```



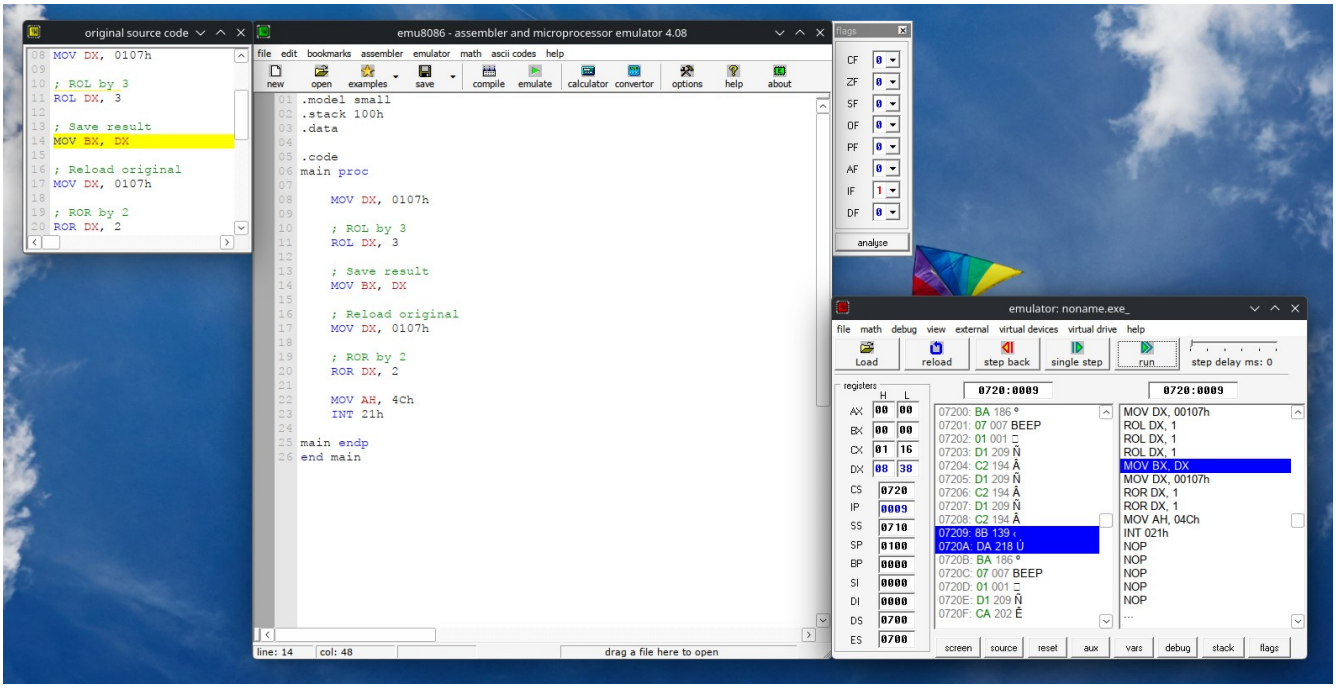
Results:

AX = 0107h = 0000 0001 0000 0111

BX = 7010h = 0111 0000 0001 0000

C. Rotate

```
.model small
.stack 100h
.data

.code
main proc

    MOV DX, 0107h

    ; ROL by 3
    ROL DX, 3

    ; Save result
    MOV BX, DX

    ; Reload original
    MOV DX, 0107h

    ; ROR by 2
    ROR DX, 2

    MOV AH, 4Ch
    INT 21h

main endp
end main
```


original source code
MOV DX, 0107h
; ROL by 3
ROL DX, 3
; Save result
MOV BX, DX
; Reload original
MOV DX, 0107h
; ROR by 2
ROR DX, 2
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file edit bookmarks assembler emulator math ascii codes help
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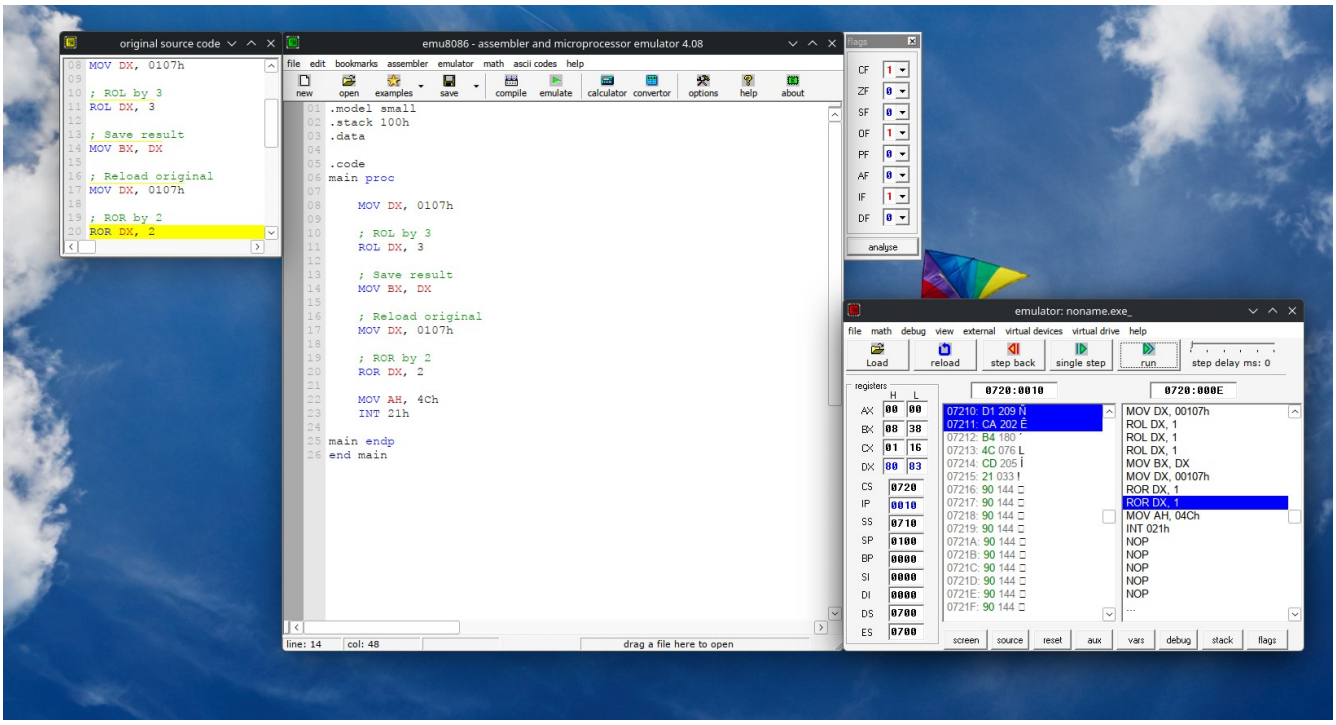
.model small
.stack 100h
.data
.code
main proc
MOV DX, 0107h
; ROL by 3
ROL DX, 3
; Save result
MOV BX, DX
; Reload original
MOV DX, 0107h
; ROR by 2
ROR DX, 2
MOV AH, 4Ch
INT 21h
main endp
end main
line: 14
col: 48
drag a file here to open
flags
CF 0
ZF 0
SF 0
OF 0
PF 0
AF 0
IF 1
DF 0
analyse
emulator: noname.exe_
file math debug view external virtual devices virtual drive help
Load reload step back single step run step delay ms: 0
registers
H L
AX 00 00
BX 00 00
CX 01 16
DX 08 38
CS 0720
IP 0009
SS 0710
SP 0100
BP 0000
SI 0000
DI 0000
DS 0700
ES 0700
0720:0009
07200: BA 186 °
07201: 07 007 BEEP
07202: 01 001
07203: D1 209 Ñ
07204: C2 194 Â
07205: D1 209 Ñ
07206: C2 194 Â
07207: D1 209 Ñ
07208: C2 194 Â
07209: 8B 139
0720A: DA 218 Ú
0720B: BA 186 °
0720C: 07 007 BEEP
0720D: 01 001
0720E: D1 209 Ñ
0720F: CA 202 Ê
0720:0009
MOV DX, 00107h
ROL DX, 1
ROL DX, 1
ROL DX, 1
MOV BX, DX
MOV DX, 00107h
ROR DX, 1
ROR DX, 1
MOV AH, 04Ch
INT 021h
NOP
NOP
NOP
NOP
NOP
screen source reset aux vars debug stack flags


original source code
MOV DX, 0107h
; ROL by 3
ROL DX, 3
; Save result
MOV BX, DX
; Reload original
MOV DX, 0107h
; ROR by 2
ROR DX, 2
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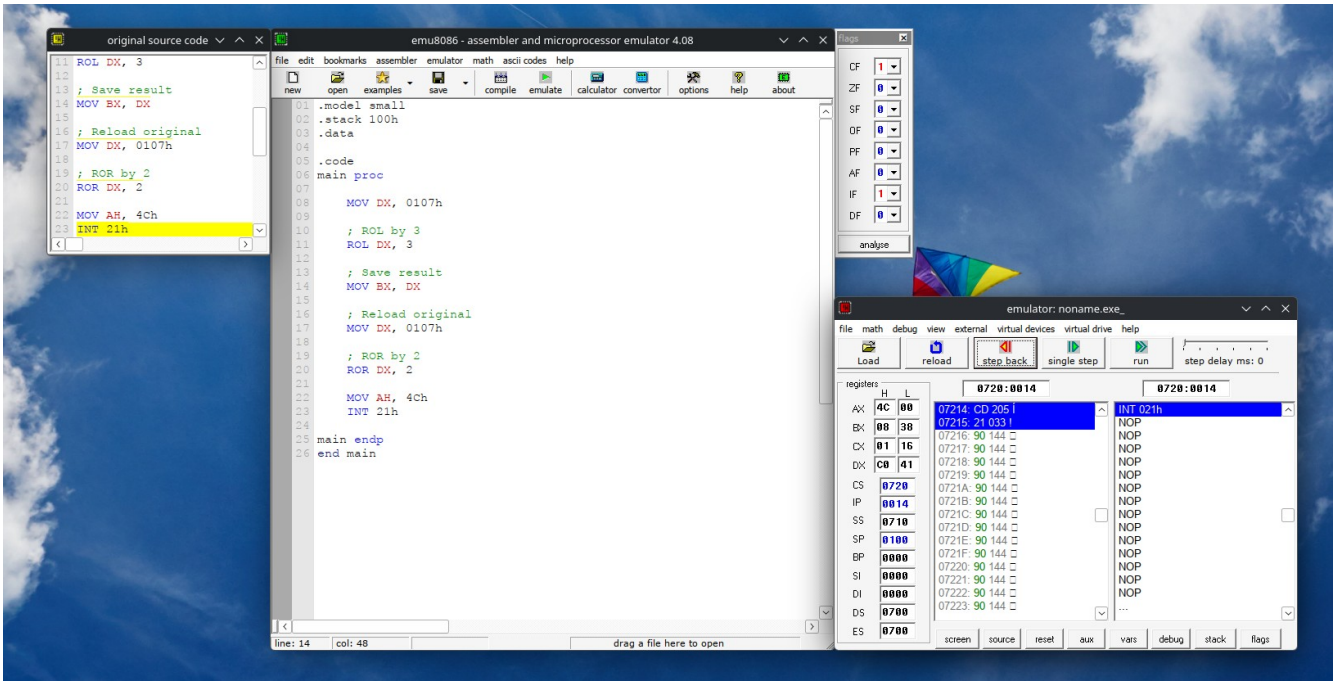
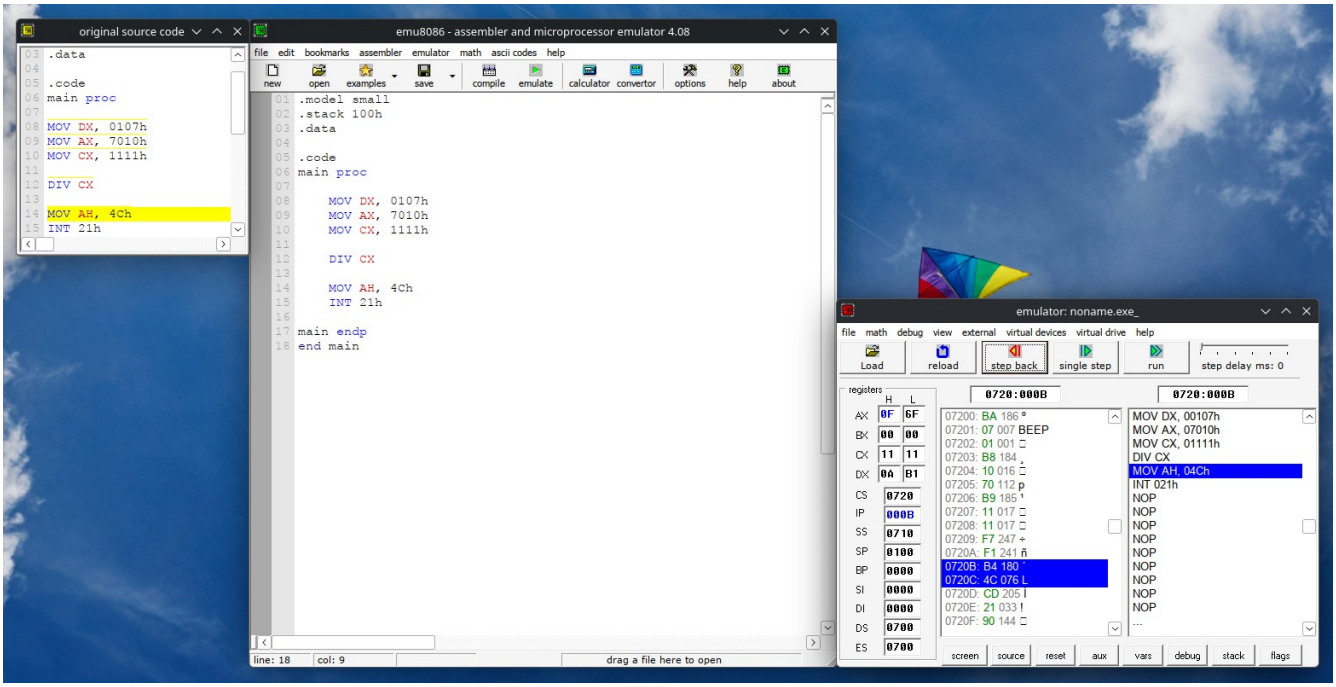
new open examples save compile emulate calculator convertor options help about
.model small
.stack 100h
.data
.code
main proc
MOV DX, 0107h
; ROL by 3
ROL DX, 3
; Save result
MOV BX, DX
; Reload original
MOV DX, 0107h
; ROR by 2
ROR DX, 2
MOV AH, 4Ch
INT 21h
main endp
end main
line: 14
col: 48
drag a file here to open
flags
CF 1
ZF 0
SF 0
OF 1
PF 0
AF 0
IF 1
DF 0
analyse
emulator: noname.exe_
file math debug view external virtual devices virtual drive help
Load reload step back single step run step delay ms: 0
registers
H L
AX 00 00
BX 08 38
CX 01 16
DX 00 03
CS 0720
IP 0010
SS 0710
SP 0100
BP 0000
SI 0000
DI 0000
DS 0700
ES 0700
0720:0010
07210: D1 209 Ñ
07211: CA 202 Ê
07212: B4 180
07213: 4C 076 L
07214: CD 205 Í
07215: 21 033 !
07216: 90 144
07217: 90 144
07218: 90 144
07219: 90 144
0721A: 90 144
0721B: 90 144
0721C: 90 144
0721D: 90 144
0721E: 90 144
0721F: 90 144
0720:000E
MOV DX, 00107h
ROL DX, 1
ROL DX, 1
ROL DX, 1
MOV BX, DX
MOV DX, 00107h
ROR DX, 1
ROR DX, 1
MOV AH, 04Ch
INT 021h
NOP
NOP
NOP
NOP
NOP
screen source reset aux vars debug stack flags



D. Division

```
.model small
.stack 100h
.data

.code
main proc

    MOV DX, 0107h
    MOV AX, 7010h
    MOV CX, 1111h

    DIV CX

    MOV AH, 4Ch
    INT 21h

main endp
end main
```



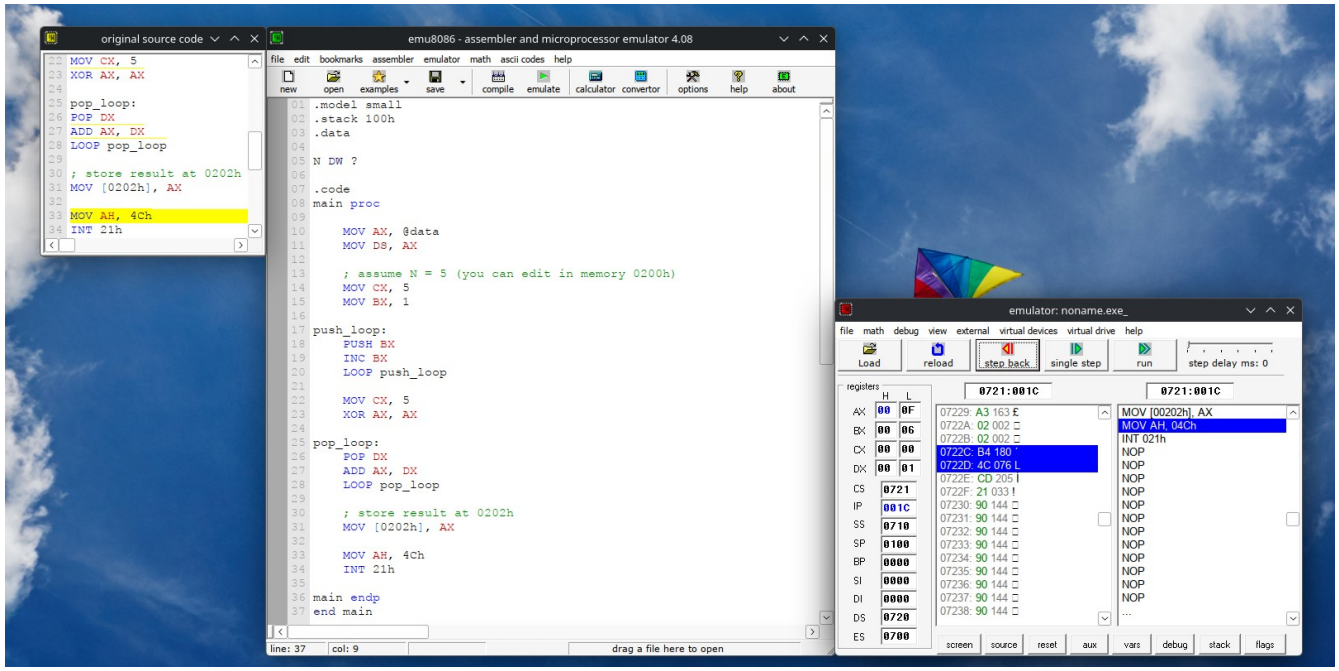
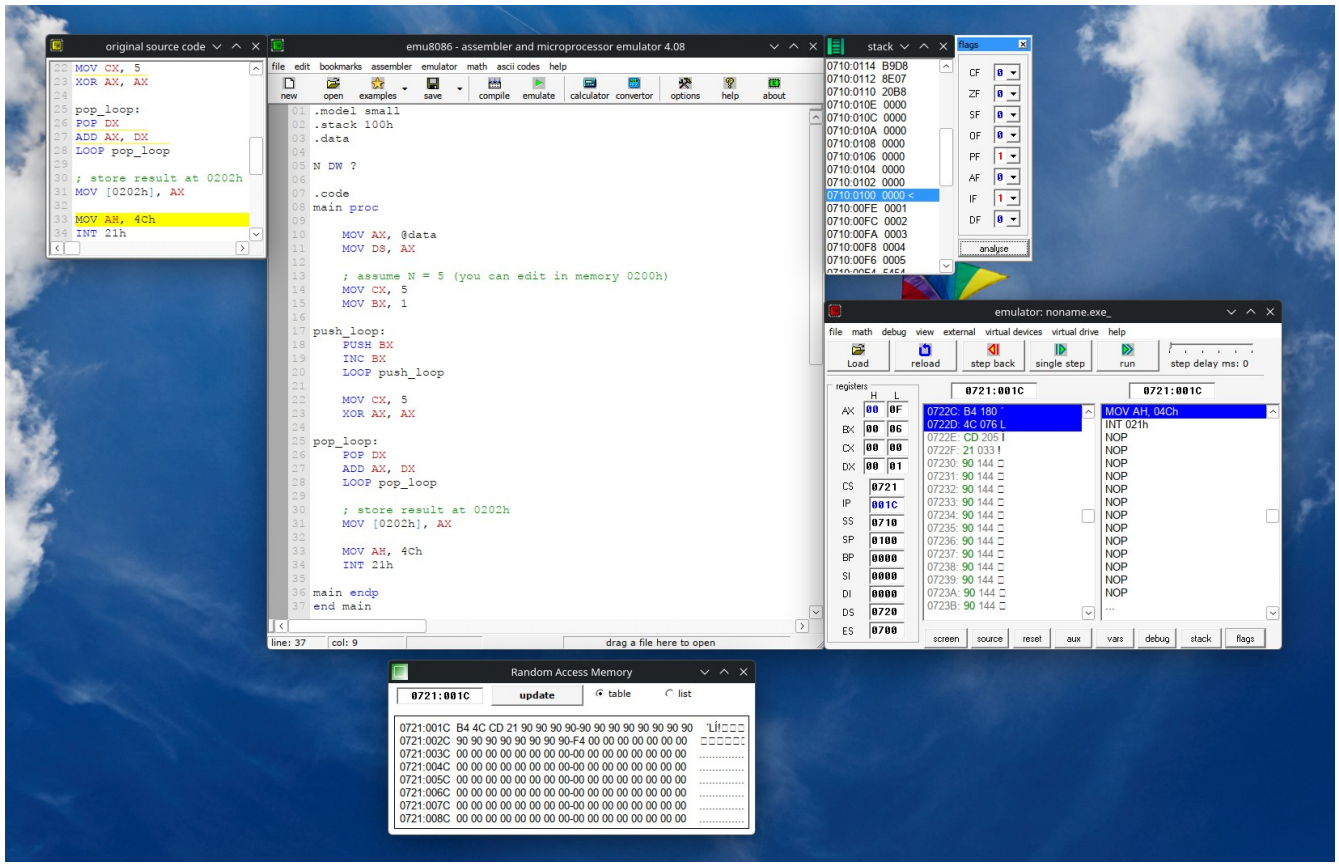
Result:

- **Quotient = 3951 = 0F6Fh**
- **Remainder = 2125 = 084Dh**

E. Sum using Stack + Loop

```
.model small

.stack 100h

.data

.code

main proc


    MOV AX, @data
```

```
    MOV DS, AX

    ; assume N = 5
    MOV CX, 5
    MOV BX, 1

push_loop:
    PUSH BX
    INC BX
    LOOP push_loop

    MOV CX, 5
    XOR AX, AX

pop_loop:
    POP DX
    ADD AX, DX
    LOOP pop_loop

    ; store result at 0202h
    MOV [0202h], AX

    MOV AH, 4Ch
    INT 21h
```

main endp

end main